



UNIVERSITÀ
degli STUDI
di CATANIA

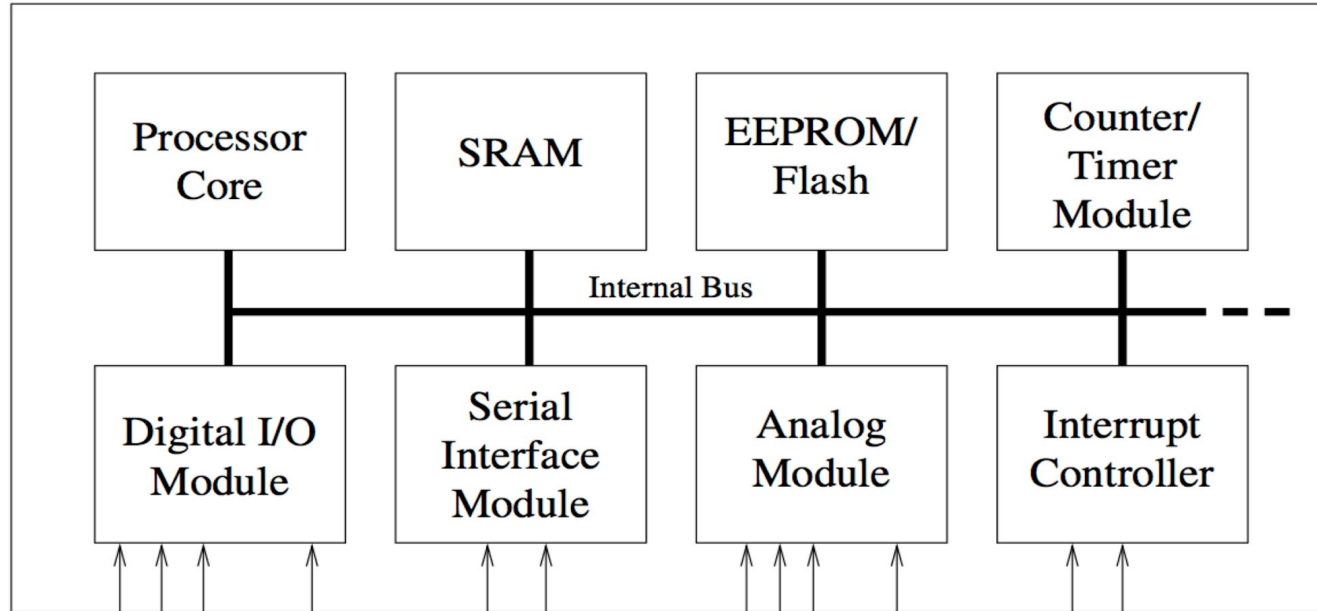
IOT Systems and Technologies

Digital and Analog I/O

davide.patti@dieei.unict.it

Basic Layout of a Microcontroller

Microcontroller

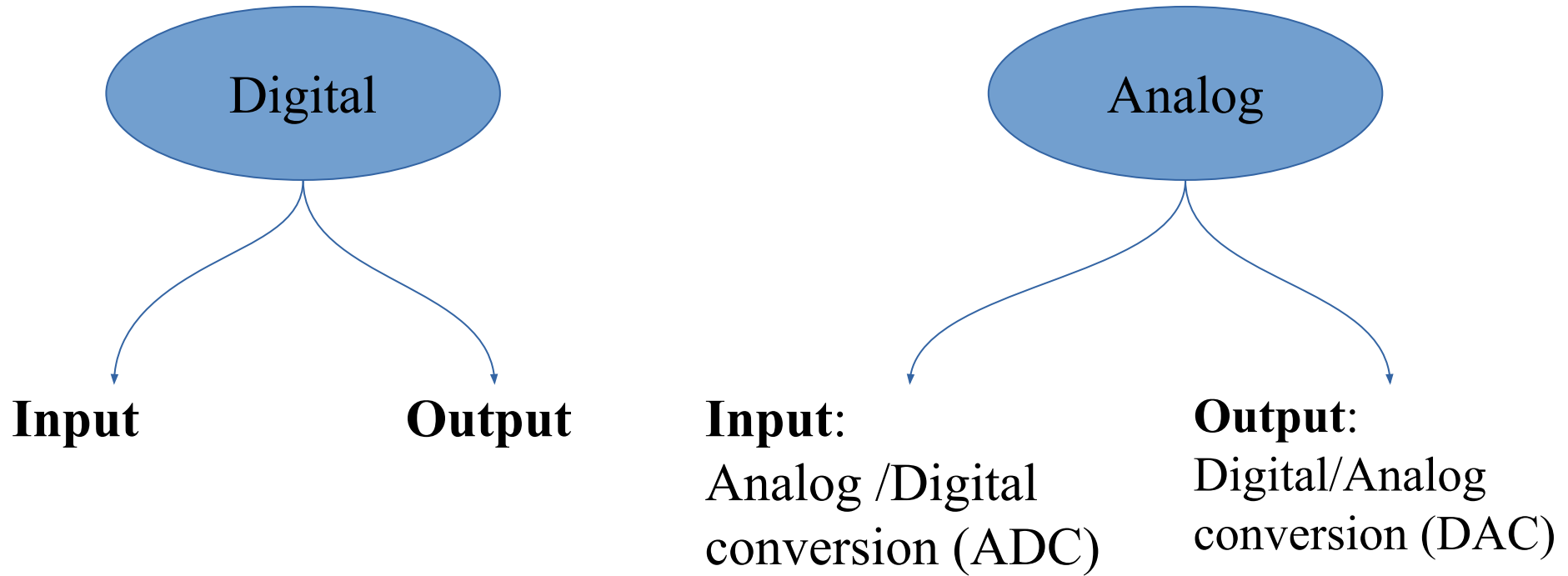


I/O pins for connecting to the external world

Embedded Systems in IoT Devices

- **Microcontroller:** the computational element of the Embedded System
- Microcontroller needs to interact with the physical world:
 - Input: Sensors (physical world->MC)
 - Output: Actuators (MC->physical world)
- **ES = Microcontroller+Sensors+Actuators**

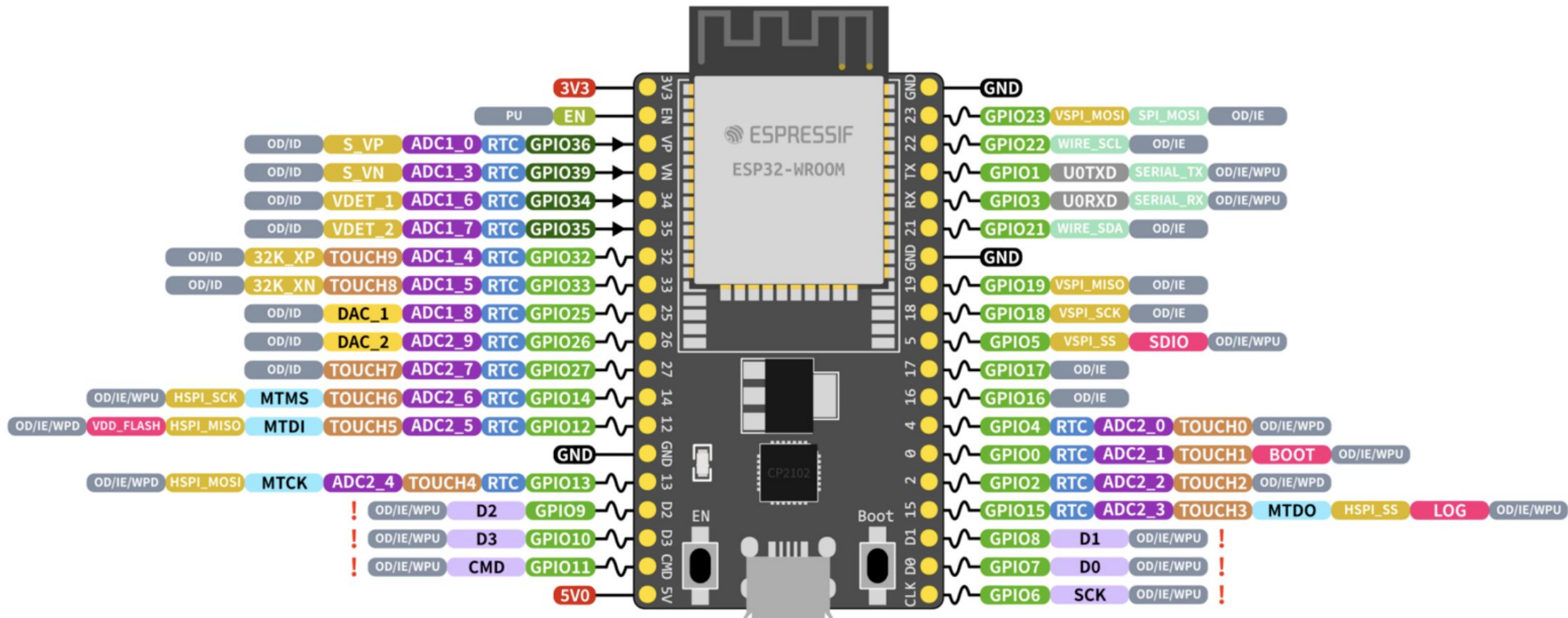




Input = “to the microcontroller”, **output** = “from the microcontroller”
Input pins are connected to sensors, Output pins to actuators

I/O Pins

- The ability to directly monitor and control hardware
 - The main characteristic of microcontrollers
- Practically all microcontrollers have at least 1-2 digital I/O pins that can be directly connected to hardware
 - In general 8-32 pins on most controllers
 - Some even have a lot more than that



ESP32 Specs

32-bit Xtensa® dual-core @240MHz

Wi-Fi IEEE 802.11 b/g/n 2.4GHz

Bluetooth 4.2 BR/EDR and BLE

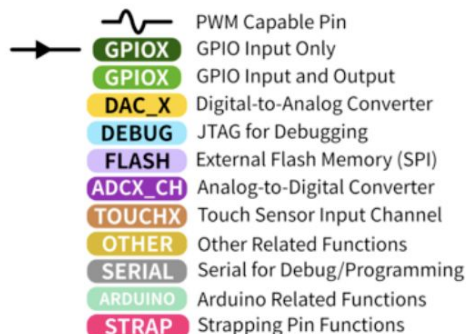
520 KB SRAM (16 KB for cache)

448 KB ROM

34 GPIOs, 4x SPI, 3x UART, 2x I2C,

2x I2S, RMT, LED PWM, 1 host SD/eMMC/SDIO,

1 slave SDIO/SPI, TWAI®, 12-bit ADC, Ethernet



RTC RTC Power Domain (VDD3P3_RTC)
GND Ground
PWD Power Rails (3V3 and 5V)
! Pin Shared with the Flash Memory
 Can't be used as regular GPIO

GPIO STATE

- WPU:** Weak Pull-up (Internal)
- WPD:** Weak Pull-down (Internal)
- PU:** Pull-up (External)
- IE:** Input Enable (After Reset)
- ID:** Input Disabled (After Reset)
- OE:** Output Enable (After Reset)
- OD:** Output Disabled (After Reset)

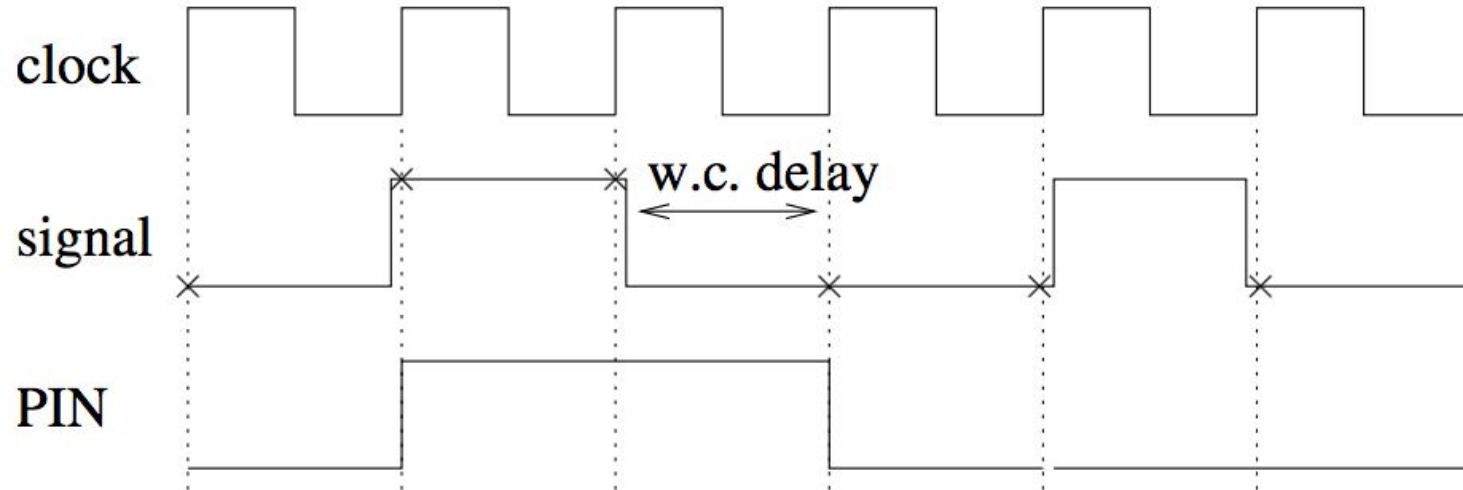
Digital Input

Digital Input

- The digital input functionality is used when the monitored signal should be interpreted as only changing between the two states:
 - High corresponding to logic 1
 - Low corresponding to logic 0
- The ESP32 recognizes a voltage above $\sim 2.0\text{V}$ (typically 2.0V to 3.3V) as HIGH.
- The ESP32 considers a voltage below $\sim 0.8\text{V}$ (typically 0V to 0.8V) as LOW.
Remaining range is undetermined

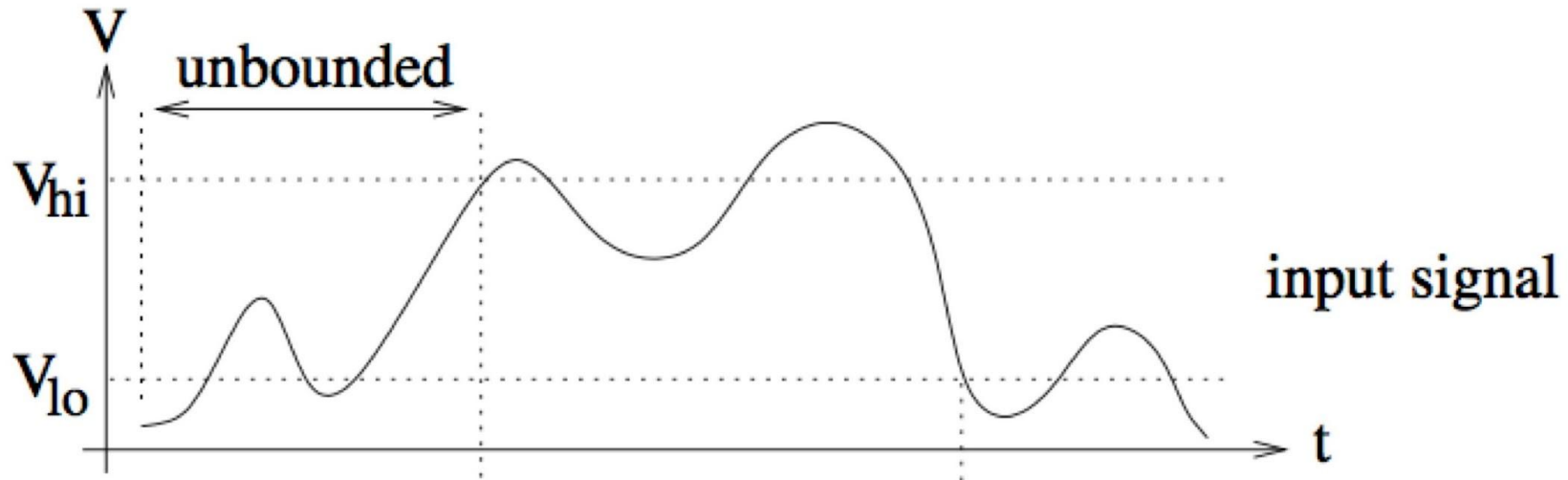
Digital Sampling

- Use latches for the PIN register and latch the current state of the pin into the register
- The latch is triggered by the system clock
 - It will store the current state at the beginning of every cycle



Undetermined Region

- The signal might be undefined when we try to latch, since the intermediate region is not deterministically associated to a specific level
- There's no guarantee “how long” the signal will stay there



Meta-stability

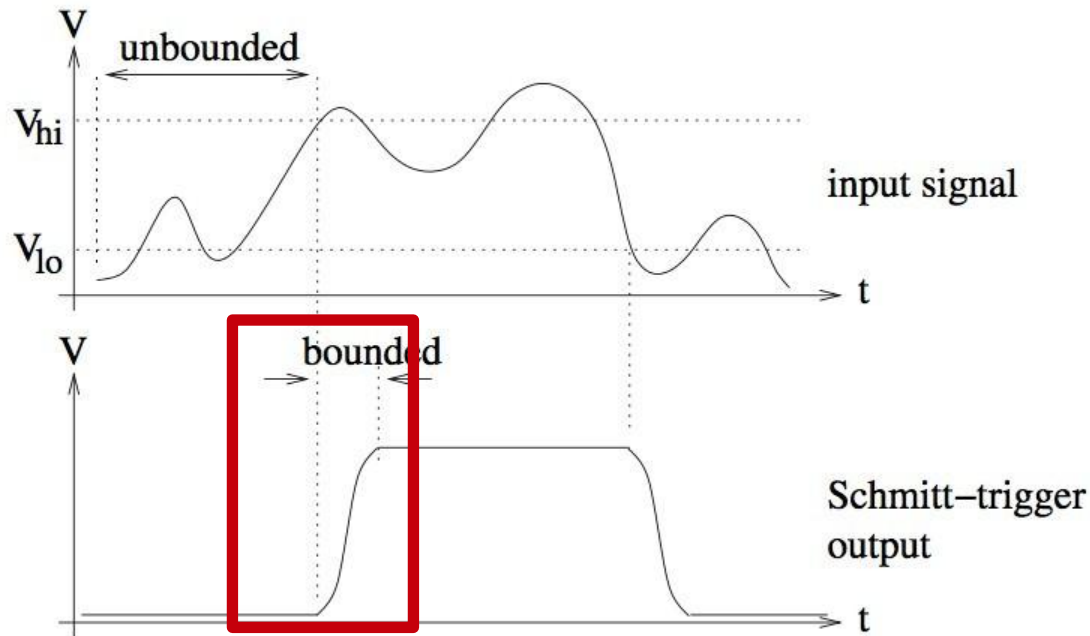
A latch that gets an undefined voltage level as input has a certain **probability p** to enter and remain in a meta-stable state

Meta-stable state: It may output either high, or low, or an undefined value, or oscillate

Obviously, the last two options are disastrous for the controller and hence for the application and must be avoided, especially in safety-critical systems.

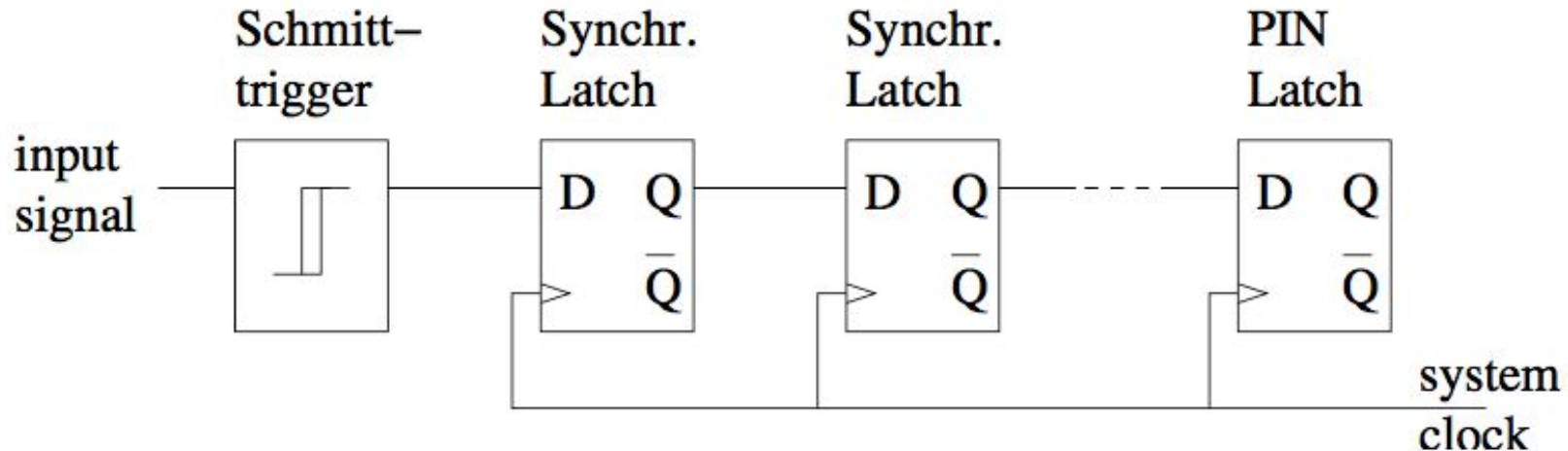
Schmitt-trigger

- Decrease the probability of meta-stability, limiting deterministically the transition time
- Ignore signal noise
- Digital input uses a Schmitt-trigger in the initial stage



Synchronizer

- To reduce the probability of meta-stability even further, one or more additional latches may be set in series between the Schmitt-trigger and the PIN register latch
- Such a construct is called a synchronizer



Synchronizer

- Each additional synchronizer latch has the probability p to enter a meta-stable state if presented with an undefined input
 - The whole chain of k latches including the PIN latch has probability $p^k \ll 1$ to pass on the meta-stable
- Disadvantage of the synchronizer stage
 - It prolongs the time until a signal change is passed to the controller by a constant d_{sync}

$$d_{\text{in}} = d_{\text{latch}} + d_{\text{sync}}$$

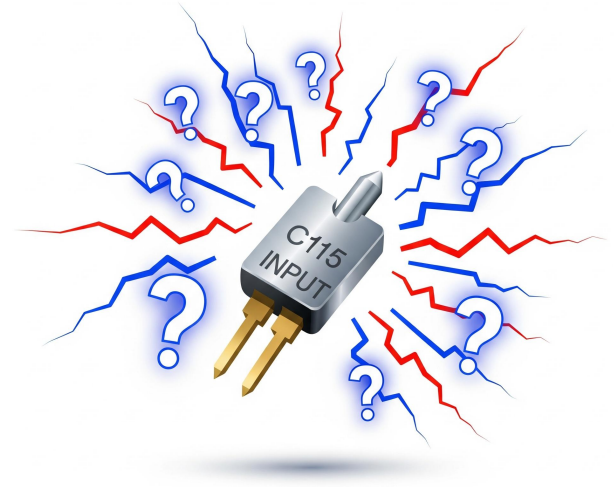
Pull Resistors

The Floating Pin Problem

A **Floating Pin** is an unconnected input that lacks a defined voltage level.

Without a pull resistor, it is highly susceptible to:

- Electromagnetic interference (noise)
- Random fluctuations between HIGH and LOW states
- Erroneous logic readings by the controller



Pull Resistors Configuration

Pull-Up Resistor (Left Diagram)

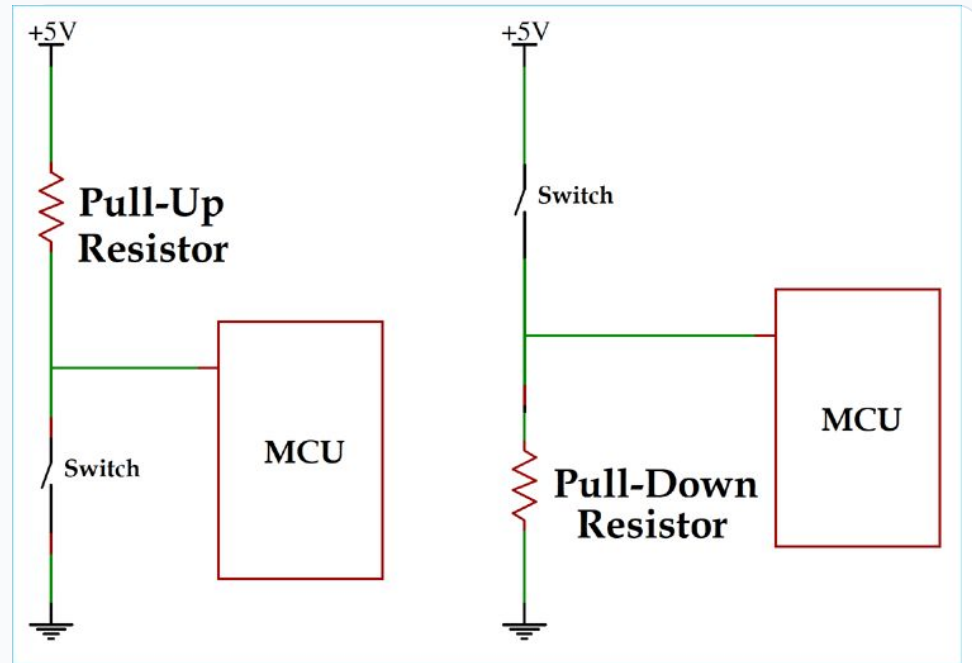
Connects the input pin to **+5V**.

- **Switch Open:** Pin is held **HIGH**.
- **Switch Pressed:** Pin is pulled **LOW** (GND).

Pull-Down Resistor (Right Diagram)

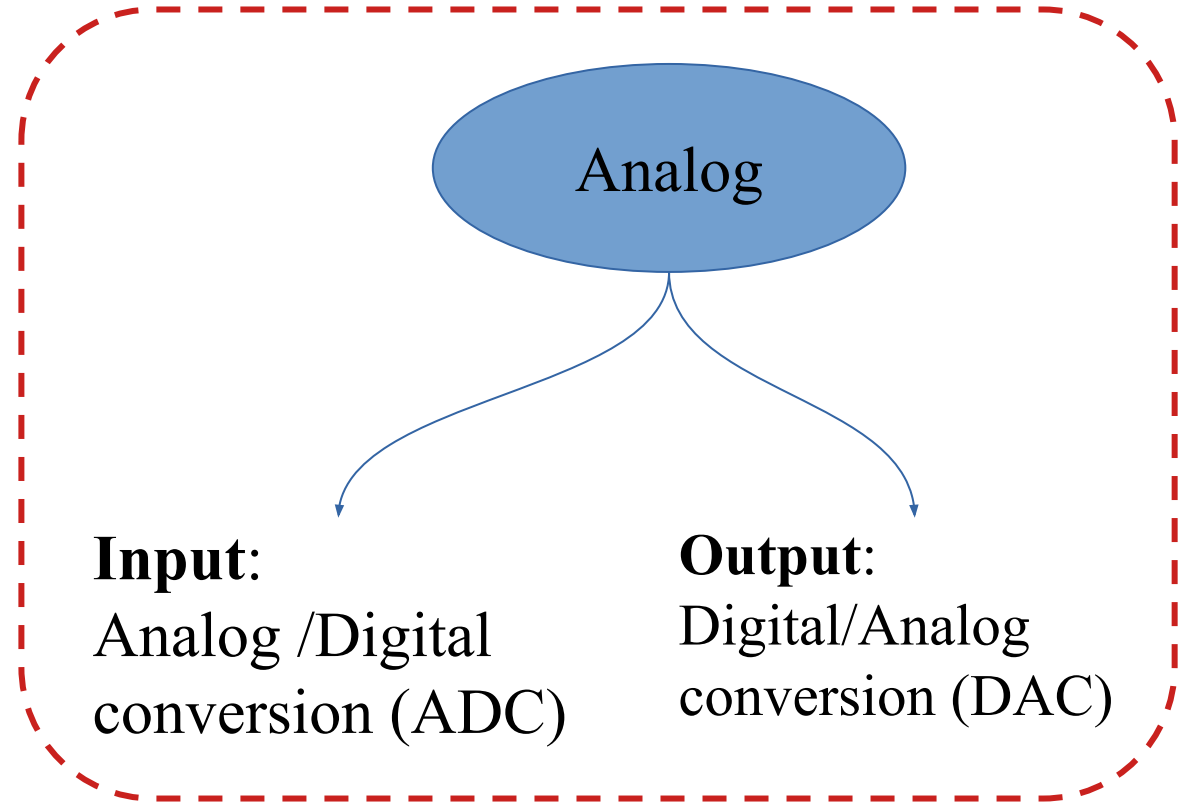
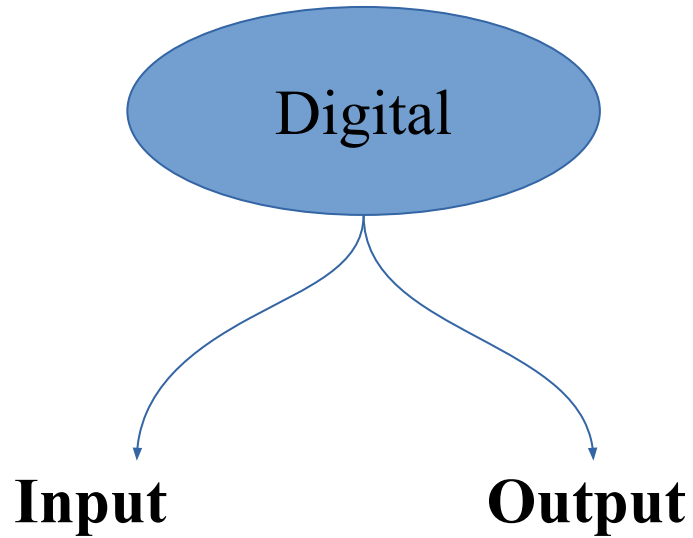
Connects the input pin to **GND**.

- **Switch Open:** Pin is held **LOW**.
- **Switch Pressed:** Pin is pulled **HIGH** (+5V).



Digital Output

- The digital output functionality is used to set output pins to given voltage levels
- The levels corresponding to high and low are again specified by the controller and depend on the controller's operating voltage



Input = “to the microcontroller”, **output** = “from the microcontroller”
Input pins are connected to sensors, Output pins to actuators

Analog I/O

In some scenarios, the **actual voltage level** on a line carries the information, rather than just binary states.



Example: Photo Transistor Light Sensor

- The output **voltage drop** is directly proportional to the amount of light detected by the transistor.
- To accurately process this data, the **microcontroller** must interpret and work with varying analog values.

Analog I/O Essentials

Microcontrollers are **inherently digital**, operating on discrete binary states.

The Bridge Challenge

Bridging the gap between the continuous analog world and digital processing is critical.

Requires efficient methods for:

- Analog-to-Digital Conversion
- Digital-to-Analog Conversion

Integrated Solution

Modern microcontrollers resolve this complexity through a specialized **Analog Module**.

This hardware peripheral manages:

- Precision signal acquisition
- Hardware-level data conversion
- Input/Output signal conditioning

Digital-to-Analog Conversion

❗ The Challenge

Objective:

- Process an r -bit digital value within the range $[0, 2^r-1]$.
- Convert this into a proportional analog output voltage (V_o).

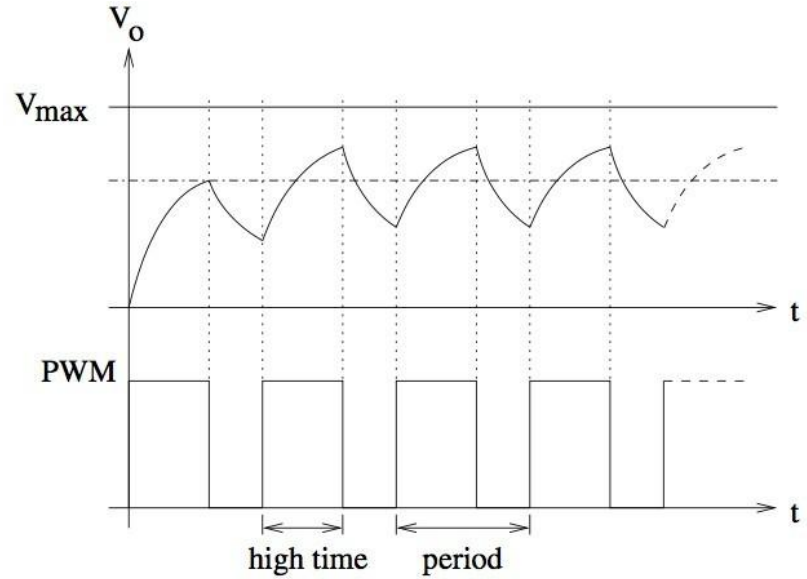
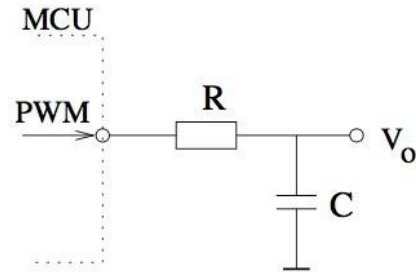
🔌 Efficient Solution: PWM

PWM + RC Low-Pass Filter:

- Simple 1-bit D/A conversion method.
- Utilizes Pulse-Width Modulation in conjunction with a passive filter.
- Highly accessible: most modern microcontrollers provide dedicated PWM outputs.

Bridging the gap between discrete binary states and the continuous analog world.

DAC Using PWM



✓ Advantages

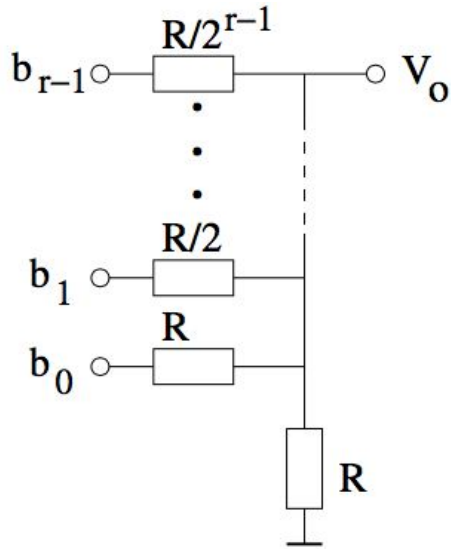
- Efficient pin usage: uses only **one single output pin**.

✗ Disadvantages

- Hardware overhead: requires a **dedicated timer** peripheral.
- Computational effort to generate the PWM signal.

Binary-Weighted Resistor Circuit

Circuit Diagram & Formula



$$V_o = V_{\text{ref}} \cdot \sum_{i=1}^r \frac{1}{2^i} b_{r-i},$$

⚠ Main Disadvantage

The circuit requires multiple resistor types that must be **high precision**.

This is essential to maintain the correct binary ratio across all stages.

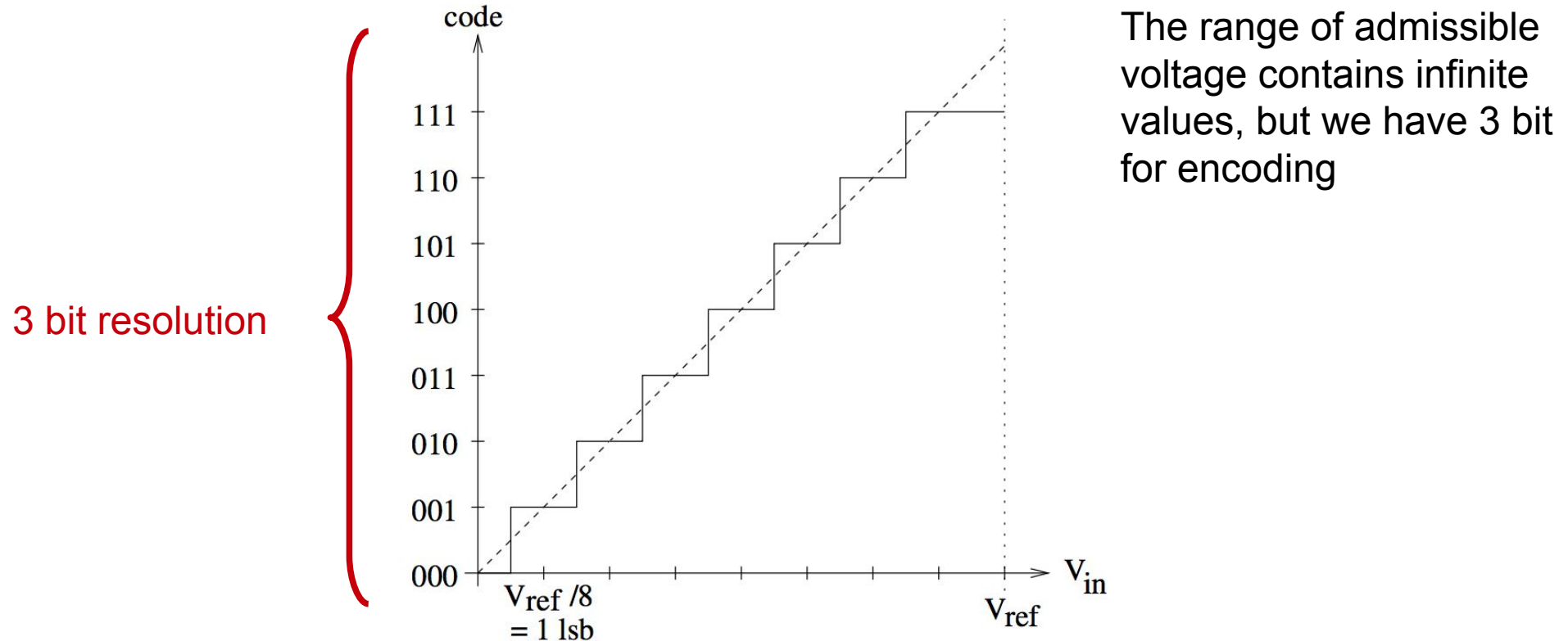
Precision requirements scale exponentially with the number of bits.

Analog/Digital Conversion

- Many microcontrollers include an analog-to-digital converter (ADC) which converts an analog input value to a binary value

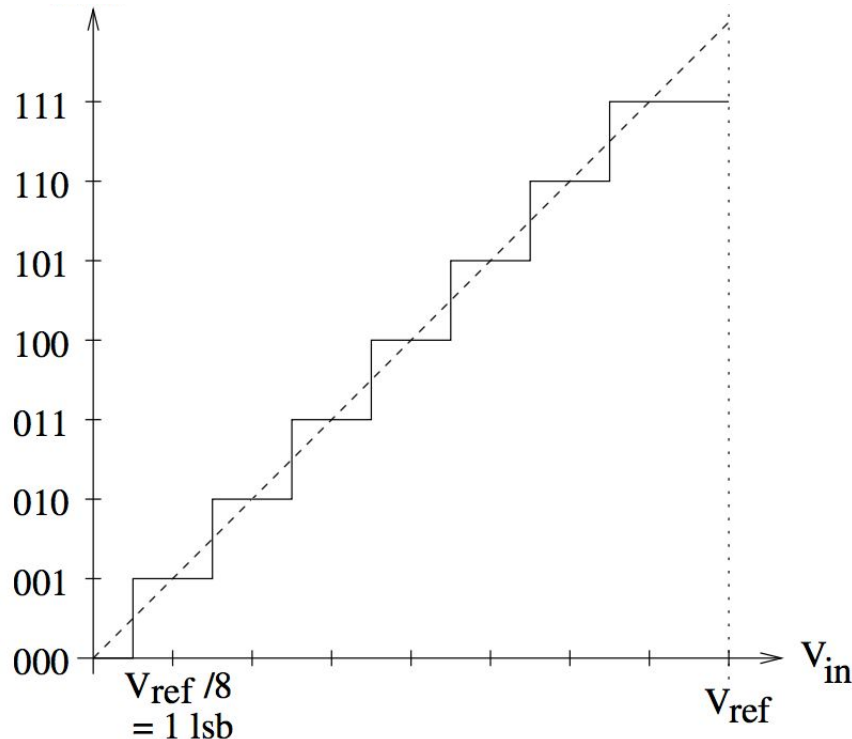


Mapping Analog Voltage to Code Words



ADC: Mapping Analog to Digital

3-Bit Quantization Example



Key Concepts

Resolution: 3 Bits

Provides **8 discrete levels** (2^3) to represent an infinite analog range.

Quantization Error: The inevitable difference between the actual analog value and the digital word.

LSB Size: In this case, $V_{ref} / 8$.

The range of admissible voltage contains infinite values, but we use a finite set of code words for encoding.

Input & Conversion Inaccuracies

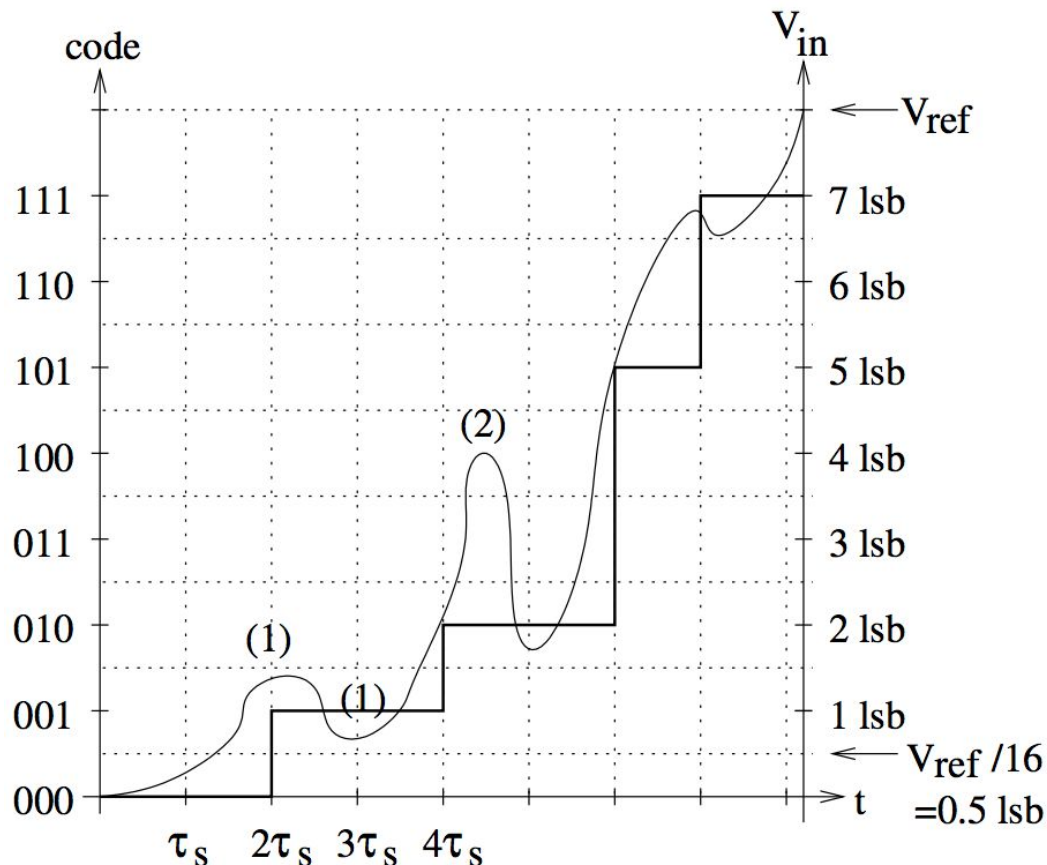
Common ADC Problems

(1) Resolution Limits

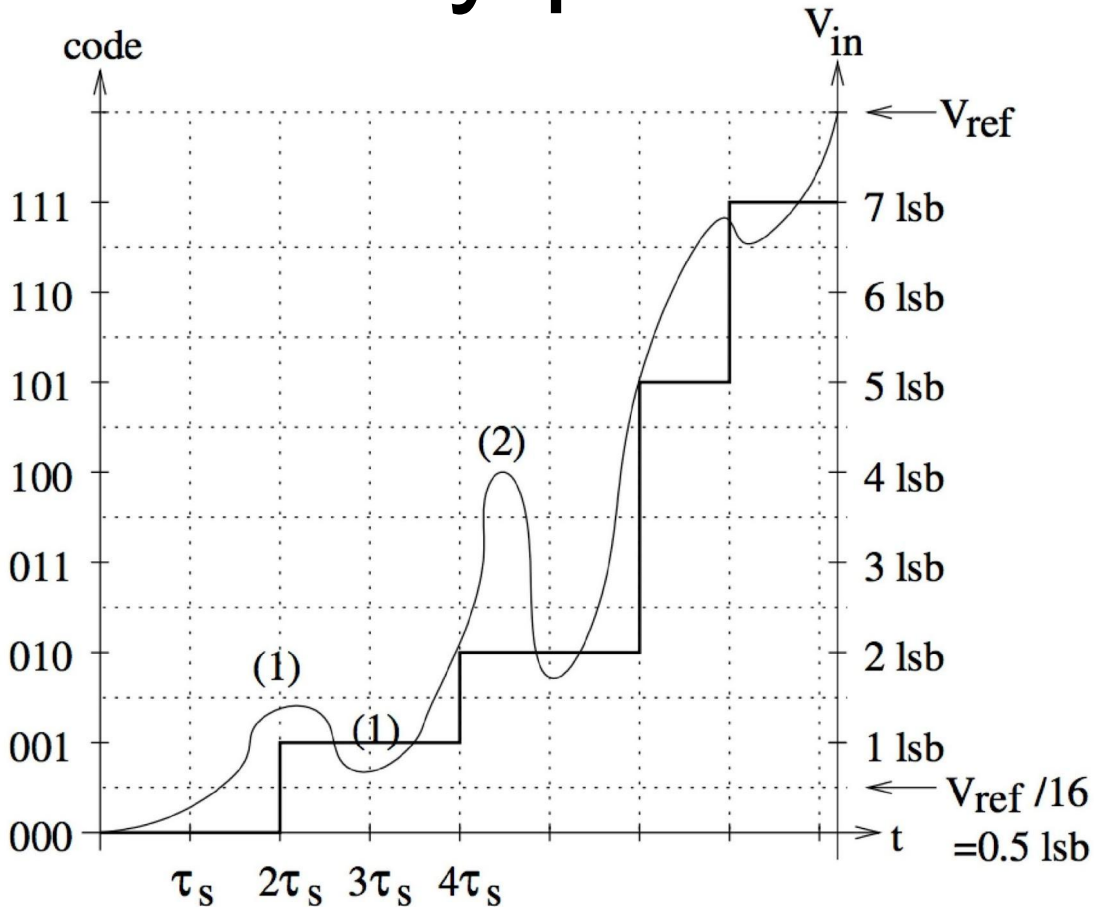
Missing signals due to insufficient quantization levels. Small variations are lost between digital steps.

(2) Sampling Frequency

Missing signals due to low sampling rates. High-frequency components are not captured

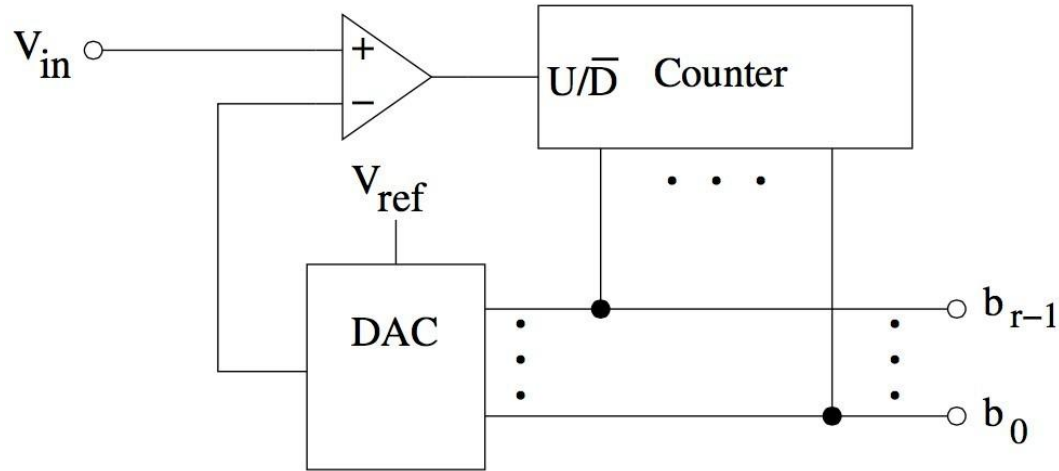


Nyquist Criterion



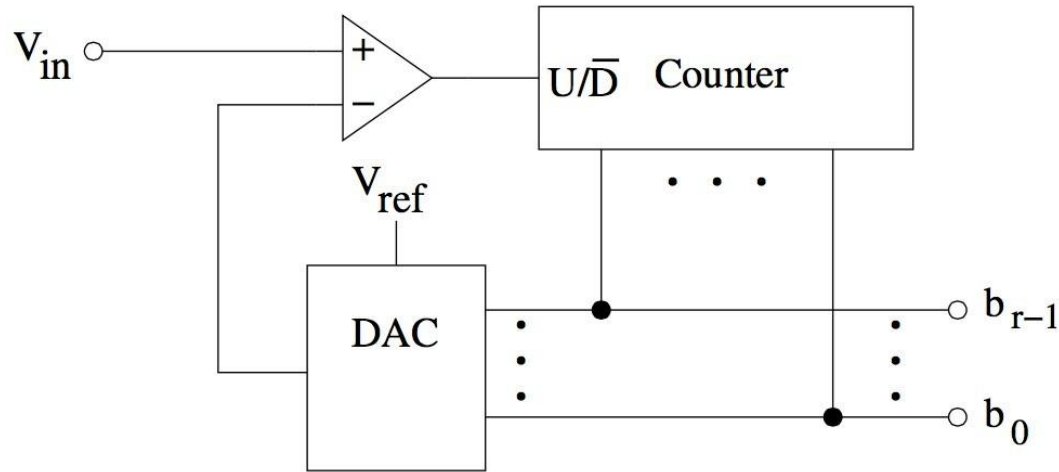
$$f_{\text{max}} < \frac{f_s}{2} = \frac{1}{2\tau_s}$$

Tracking Converter



- The heart of the tracking converter is a counter which holds the current digital estimate of the input voltage.
- The counter value is converted to an analog value by the DAC and compared to the input voltage.
- If the input voltage is greater than the current counter value, then the counter is incremented, otherwise it is decremented.

Tracking ADC Converter



Core Mechanism

The heart of the tracking converter is a counter which holds the current digital estimate of the input voltage.

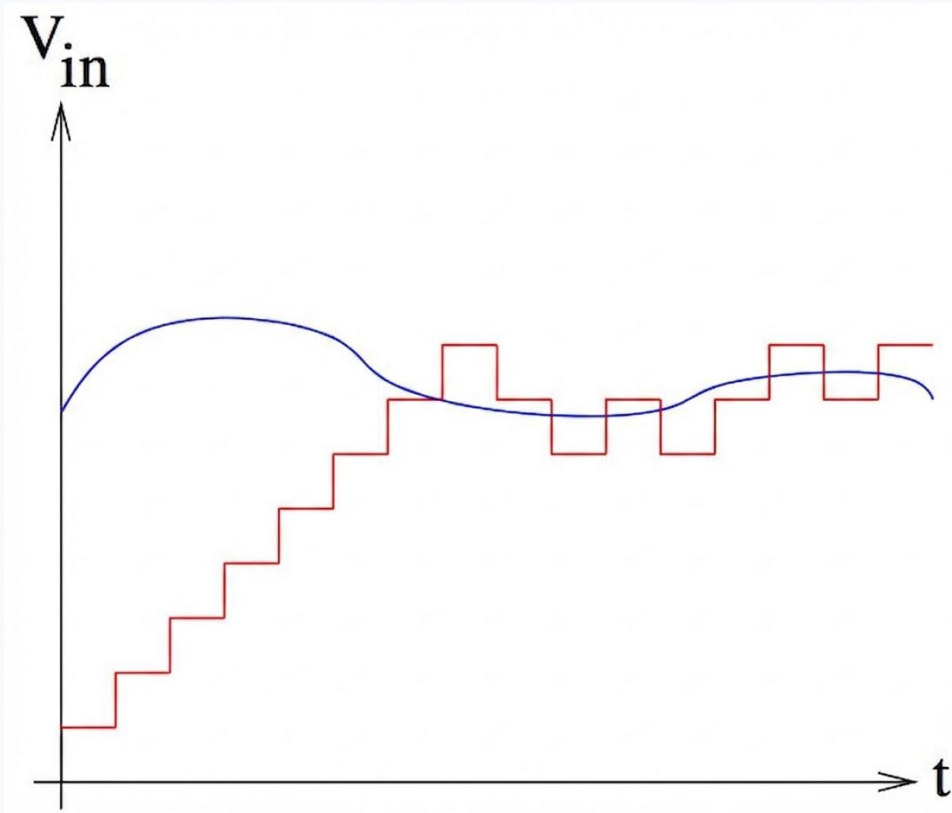
Closed-Loop Feedback

The counter value is converted to an analog value by the DAC and compared to the input voltage.

Tracking Logic

If $V_{in} > V_{ref}$, the counter increments; otherwise, it decrements to track the signal.

Tracking Converter in Action



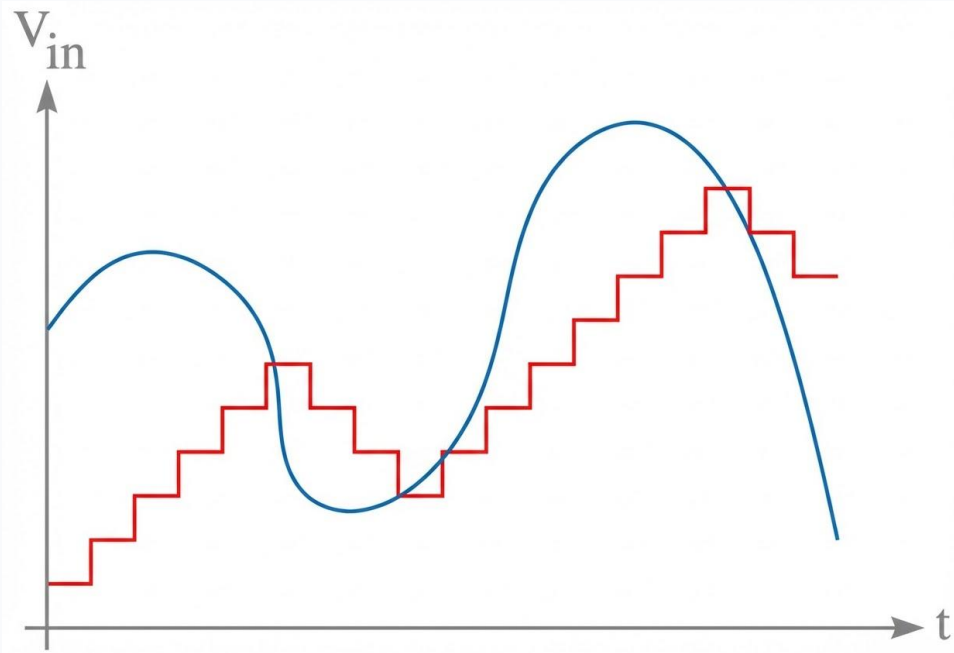
Initial Acquisition

Requires a significant amount of time to initially catch and lock onto the signal.

Steady State Tracking

Once locked, the conversion time becomes quite fast as it only tracks small signal variations.

Tracking Converter in Action



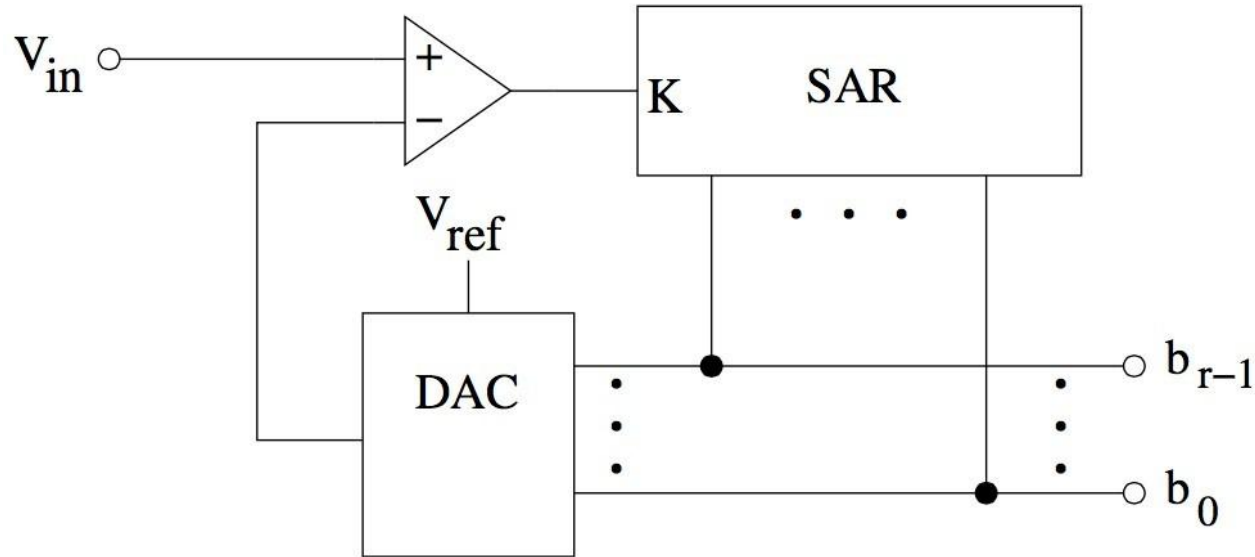
High Signal Slew Rate

If the input signal changes too rapidly for the tracking step size...

Tracking Limitations

The converter spends most of its operational time attempting to catch up with the signal.

Successive Approximation Converter



The SAR implements a binary search instead of simply counting up or down

Successive Approximation Converter

